COMMUNICATION ELECTRONICS
2nd Edition by *LOUIS E. FRENZEL*
Chapter 1: Introduction to Electronic Communications

1. Communication is the process of
    a. Keeping in touch
    b. Broadcasting
    c. Exchanging Information
    d. Entertainment by Electronics

2. Two key barriers to human communication are
    a. Distance
    b. Cost
    c. Ignorance
    d. Language

3. Electronic Communications is discovered on which century?
    a. Sixteenth
    b. Eighteenth
    c. Nineteenth
    d. Twentieth

4. Which of the following is not a major communication medium?
    a. Free Space
    b. Water
    c. Wires
    d. Fiber Optic Cable

5. Random interference to transmitted signal is called
    a. Adjacent channel overlap
    b. Crosstalk
    c. Garbage in – Garbage out
    d. Noise

6. The communications medium causes the signal to be
    a. Amplified
    b. Modulated
    c. **Attenuated**
    d. Interferred with

7. Which of the following is not a source of noise?
    a. Another communication signal
    b. Atmospheric effects
    c. Manufactured electrical systems
    d. Thermal agitation in electronic components

8. One way communication is called
    a. Half Duplex
    b. Full Duplex
    c. Monocomm.
    d. Simplex

9. Simultaneous two way communication is called
    a. Half Duplex
    b. Full Duplex
    c. Bicomm.
    d. Simplex

10. The original electrical information signal to be transmitted is called the
    a. Modulating signal
    b. Carrier
    c. Baseband signal
    d. Source signal

11. The process of modifying a high-frequency carrier with the information to be transmitted is called the
    a. Multiplexing
    b. Telemetry
    c. Detection
    d. Modulation

12. The process of transmitting two or more information signals simultaneously over the same channel is called
    a. Multiplexing
    b. Telemetry
    c. Mixing
    d. Modulation

13. Continuous voice or video signals are referred to as being
    a. Baseband
    b. Analog
    c. Digital
    d. Continuous waves

14. Recovering information from a carrier is known as
    a. demultiplexing
    b. modulation
    c. detection
    d. carrier recovery

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15. Transmission of graphical information over the telephone network is accomplished by
    a. Television
    b. CATV
    c. Videotext
    d. Facsimile

16. Measuring physical conditions at some remote location and transmitting this data for analysis is the process of
    a. Telemetry
    b. Instrumentation
    c. Modulation
    d. Multiplexing

17. Receiving electromagnetic emissions from stars is called
    a. Astrology
    b. Optical astronomy
    c. Radio Astronomy
    d. Space surveillance

18. A personal communications hobby for individuals is
    a. Ham radio
    b. Electronic Bulletin Board
    c. CB radio
    d. Cellular radio

19. Radar is based upon
    a. Microwaves
    b. A water medium
    c. The directional nature of radio signals
    d. Reflected radio signals

20. A frequency of 27MHz has a wavelength of approximately
    a. 11 m
    b. 27 m
    c. 30 m
    d. 81 m

21. Radio signals are made of
    a. Voltages and Currents
    b. Electric and Magnetic Fields
    c. Electrons and protons
    d. Noise and data

22. The voice frequency range is
    a. 30 to 300Hz
    b. 300 to 3000Hz
    c. 20Hz to 20KHz
    d. 0Hz to 15Hz

23. Another name for signals in the HF range is
    a. Microwaves
    b. RF waves
    c. Short waves
    d. Millimeter waves

24. Television broadcasting occurs in which range
    a. HF
    b. EHF
    c. VHF
    d. UHF

25. Electromagnetic waves produced primarily by heat are called
    a. Infrared rays
    b. Microwaves
    c. Shortwaves
    d. X-rays

26. A micron is
    a. one millionth of a foot
    b. one millionth of a meter
    c. one thousandth of a meter
    d. one ten thousandth of an inch

27. The frequency range of infrared rays is approximately
    a. 30 to 300GHZ
    b. 4000 to 8000 A
    c. 1000 to 10000 A
    d. 0.7 to 100 μm

28. The approximate wavelength of red light is
    a. 1000 μm
    b. 7000 A
    c. 3500 A
    d. 4000 A

29. Which of the following is not used for communications?
    - **a. X-rays**
    - b. Millimeter waves
    - c. infrared
    - d. microwaves

30. A signal occupies the spectrum space from 1.115 to 1.122GHz. the bandwidth is
    - a. 0.007 MHz
    - **b. 7 MHz**
    - c. 237 MHz
    - d. 700 MHz

31. In the United States, the electromagnetic spectrum is regulated and managed by
    - a. business and industry
    - b. ITU
    - **c. FCC**
    - d. the united nations

32. For a given bandwidth signal, more channel space is available for signals in the range of
    - a. VHF
    - b. UHF
    - c. SHF
    - **d. EHF**

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Chapter 2: AM and SSB Modulation

1. Having an information signal change some characteristic of a carrier signal is called
   a. Multiplexing
   b. Modulation
   c. Duplexing
   d. Linear mixing

2. Which of the following is not true about AM?
   a. the carrier amplitude varies
   b. the carrier frequency remains constant
   c. the carrier frequency changes
   d. the information signal amplitude changes the carrier amplitude

3. The opposite of modulation is
   a. reverse modulation
   b. downward modulation
   c. unmodulation
   d. demodulation

4. The circuit used to produced modulation is called a
   a. Modulator
   b. Demodulator
   c. Variable Gain amplifier
   d. Multiplexer

5. A modulator circuit performs what mathematical operation on its two inputs?
   a. addition
   b. multiplication
   c. division
   d. square root

6. The ratio of the peak modulating signal voltage to the peak carrier voltage is referred to as
   a. the voltage ratio
   b. decibels
   c. modulation index
   d. mix factor

7. If m is greater than 1, what happens?
   a. normal operation
   b. carrier frequency shifts
   c. carrier drops to zero
   d. information signal is distorted

8. For ideal AM, which of the following is true?
   a. $m = 0$
   b. $m = 1$
   c. $m < 1$
   d. $m > 1$

9. The outline of the peaks of a carrier has the shape of the modulating signal and is called the
   a. trace
   b. waveshape
   c. envelope
   d. carrier variation

10. Overmodulation occurs when
   a. $V_m > V_c$
   b. $V_m < V_c$
   c. $V_m = V_c$
   d. $V_m = V_c = 0$

11. The values of Vmax and Vmin as read from an AM wave on an oscilloscope are 2.8 and 0.3. The percentage of modulation is
   a. 10.7 percent
   b. 41.4 percent
   c. 80.6 percent
   d. 93.3 percent

12. The new signals produced by modulation are called
   a. spurious emissions.
   b. harmonics
   c. intermodulation products
   d. sidebands

13. A carrier of 880 kHz is modulated by a 3.5 kHz sine wave. The LSB and USB are, respectively,
   a. 873 and 887 kHz
   b. 876.5 and 883.5 kHz
   c. 883.5 and 876.5 kHz
   d. 887 and 873 kHz

14. A display of signal amplitude versus frequency is called the
   a. time domain
   b. frequency spectrum
   c. amplitude spectrum
   d. frequency domain

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15. Most of the power in an AM signal is the
    a. **carrier**
    b. upper sideband
    c. lower sideband
    d. modulating signal

16. An AM signal has a carrier power of 5 W. The percentage of modulation is 80 percent. The total sideband power is
    a. 0.8 W
    b. **1.6 W**
    c. 2.5 W
    d. 4.0 W

17. For 100 percent modulation, what percentage of power is in each sideband?
    a. **25 percent**
    b. 33.3 percent
    c. 50 percent
    d. 100 percent

18. An AM transmitter has a percentage of modulation of 88. The carrier power is 440 W. The power in one sideband is
    a. **85 W**
    b. 110 W
    c. 170 W
    d. 610 W

19. An AM transmitter antenna current is measured with no modulation and found to be 2.6 A. With modulation, the current rises to 2.9 A. The percentage of modulation is
    a. 35 percent
    b. **70 percent**
    c. 42 percent
    d. 89 percent

20. What is the carrier power in the problem above if the antenna resistance is 75 ohms?
    a. 195 W
    b. 631 W
    c. **507 W**
    d. 792 W

21. In an AM signal, the transmitted information is contained within the
    a. carrier
    b. modulating signal
    c. **sidebands**
    d. envelope

22. An AM signal without the carrier is called a(n)
    a. SSB
    b. vestigial sideband
    c. FM signal
    d. **DSB**

23. What is the minimum AM signal needed to transmit information?
    a. carrier plus sidebands
    b. carrier only
    c. **one sideband**
    d. both isdeband

24. The main advantage of SSB over standard AM or DSB is
    a. **less spectrum space is used**
    b. simpler equipment is used
    c. less power is consumed
    d. a higher modulation percentage

25. In SSB, which sideband is the best to use?
    a. upper
    b. lower
    c. **neither**
    d. depends upon use

26. The typical audio modulating frequency range used in radio and telephone communications is
    a. 50 Hz to 5KHz
    b. 50 Hz to 15 Khz
    c. 100 Hz to 10kHz
    d. **300hz to 3khz**

27. An AM signal with a maximum modulating signal frequency of 4.5 kHz has a total bandwidth of
    a. 4.5 kHz
    b. 6.75 kHz
    c. **9 Khz**
    d. 18 kHz

28. Distortion of the modulating signal produces harmonics which cause an increase in the signal
    a. carrier power
    b. **bandwidth**
    c. sideband power
    d. envelope voltage

29. The process of translating a signal, with or without modulation, to a higher or lower frequency for processing is called
    a. frequency multiplication
    b. frequency division
    c. frequency shift
    d. frequency conversion

30. Frequency translation is carried out by a circuit called a
    a. translator
    b. converter
    c. balanced modulator
    d. local oscillator

31. An input signal of 1.8 MHz is mixed with a local oscillation of 5 MHz. A filter selects the difference signal. The output is
    a. 1.8 MHz
    b. 3.2 MHz
    c. 5 MHz
    d. 6.8 MHz

32. The output of an SSB transmitter with a 3.85 MHz carrier and a 1.5 kHz sine wave modulating tone is
    a. a 3.8485 MHz sine wave
    b. a 3.85 Mhz sine wave
    c. a 3.85, 3.8485, and 3.8515 Mhz sine wave
    d. 3848.5 and 3851.5 MHz sine wave

33. An SSB transmitter produces a 400 V peak-to-peak signal across a 52 ohms antenna load. The PEP output is
    a. 192.2 W
    b. 384.5 W
    c. 769.2 W
    d. 3077 W

34. The output power of an SSB transmitter is usually expressed in terms of
    a. average power
    b. RMS power
    c. peak to peak power
    d. peak envelope power

35. An SSB transmitter has a PEP rating of 1 kW. The average output power is in the range of
    a. 150 to 450 W
    b. 100 to 300 W
    c. 250 to 333 W
    d. 3 to 4 kW

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Chapter 3: Amplitude Modulation Circuits

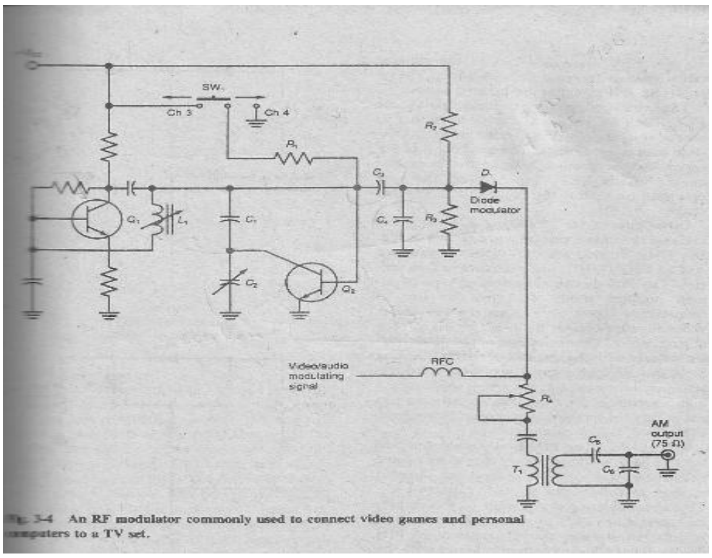
1. Amplitude modulation is the same as
   a. Linear Mixing
   b. Analog Multiplication
   c. Signal Summation
   d. Multiplexing

2. In a diode modulator, the negative half of the AM wave is supplied by a(n)
   a. Tuned Circuit
   b. Transformer
   c. Capacitor
   d. Inductor

3. Amplitude modulation can be produced by
   a. Having the carrier vary a resistance
   b. Having the modulating signal vary a capacitance
   c. Varying the carrier frequency
   d. Varying the gain of an amplifier

4. Amplitude modulators that vary the carrier amplitude with the modulating signal by passing it through an attenuator work on the principle of
   a. Rectification
   b. Resonance
   c. Variable resistance
   d. Absorption

5. In Fig. 3-4, D_1 is a
   a. Variable resistor
   b. Mixer
   c. Clipper
   d. Rexctifier



3-4 An RF modulator commonly used to connect video games and personal ...mputers to a TV set.

6. The component used to produce AM at very high frequencies is a
   a. Varactor
   b. Thermistor
   c. Cavity resonator
   d. PIN diode

7. Amplitude modulation generated at a very low voltage or power amplitude is known as
   a. High-level Modulation
   b. Low-level Modulation
   c. Collector Modulation
   d. Minimum Modulation

8. A collector modulator has a supply voltage of 48 V. The peak-to-peak amplitude of the modulating signal for 100 percent modulation is
   a. 24 V
   b. 48 V
   c. 96 V
   d. 120 V

9. A collector-modulated transmitted has a supply voltage of 24 V and a collector current of 0.5A. The modulator power for 100 percent modulation is
   a. 6 W
   b. 12 W
   c. 18 W
   d. 24 W

10. The circuit that recovers the original modulating information from an AM signal is known as a
   a. Modulator
   b. Demodulator
   c. Mixer
   d. Crystal set

11. The most commonly used amplitude demodulator is the
   a. Diode Mixer
   b. Balanced Modulator
   c. Envelope Detector
   d. Crystal Filter

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Chapter 3: Amplitude Modulation Circuits

12. A circuit that generates the upper and lower sidebands but no carriers is called a(n)
    a. Amplitude Modulator
    b. Diode Detector
    c. Class C Amplifier
    d. Balanced Modulator

13. The inputs to a balanced modulator are 1 Mhz and a carrier of 1.5 Mhz. The outputs are
    a. 500 kHz
    b. 2.5 MHz
    c. 1.5 MHz
    d. All of the above
    e. a and b

14. A widely used balanced modulator is called the
    a. Diode Bridge Circuit
    b. Full-wave Bridge Rectifier
    c. Lattice Modulator
    d. Balanced Bridge Modulator

15. In a diode ring modulator, the diodes act like
    a. Variable Resistors
    b. Switches
    c. Rectifiers
    d. Variable capacitors

16. The output of a balanced modulator is
    a. AM
    b. FM
    c. SSB
    d. DSB

17. The principal circuit in the popular 1496/1596 IC balanced modulator is a
    a. Differential Amplifier
    b. Rectifier
    c. Bridge
    d. Constant Current Source

18. The most commonly used filter in SSB generators uses
    a. LC networks
    b. Mechanical Resonators
    c. Crystals
    d. RC Networks and Op amps

19. The equivalent circuit of a quartz crystal is a
    a. Series Resonant Circuit
    b. Parallel Resonant Circuit
    c. Neither a nor b
    d. Both a and b

20. A crystal lattice filter has crystal frequencies of 27.5 and 27.502 MHz. The bandwidth is approximately
    a. 2 kHz
    b. 3 kHz
    c. 27.501 MHz
    d. 55.502 MHz

21. An SSB generator has a sideband filter centered at 3.0 MHz. The modulating signal is 3 kHz. To produce both upper and lower sidebands, the following carrier frequencies must be produced:
    a. 2.7 and 3.3 MHz
    b. 3.3 and 3.6 MHz
    c. 2997 and 3003 kHz
    d. 3000 and 3003 kHz

22. In the phasing method of SSB generation, one sideband is canceled out due to
    a. Phase Shift
    b. Sharp Selectivity
    c. Carrier Suppression
    d. Phase Inversion

23. A balanced modulator used to demodulate a SSB signal is called a(n)
    a. Transponder
    b. Product Detector
    c. Converter
    d. Modulator

24. Frequency translation is done with circuit called a
    a. Summer
    b. Multiplier
    c. Filter
    d. Mixer

25. The inputs to a mixer are f_o and f_m. In down conversion, which of the following mixer output signals is selected?
    a. f_o
    b. f_m
    c. f_o - f_m
    d. $f_o + f_m$

26. Mixing for frequency conversion is the same as
    a. Rectification
    b. AM
    c. Linear Summing
    d. Filtering

27. Which of the following can be used as a mixer
    a. Balanced Modulator
    b. FET
    c. Diode Modulator
    d. All of the above

28. The desired output from a mixer is usually selected with a
    a. Phase-shift circuit
    b. Crystal Filter
    c. Resonant Circuit
    d. Transformer

29. The two inputs to a mixer are the signal to be translated and a signal from a(n)
    a. Modulator
    b. Filter
    c. Antenna
    d. Local Oscillator

30. An NE602 mixer IC has a difference output of 10.7 MHz. The input is 146.8 MHz. The local oscillator frequency is
    a. 101.9 MHz
    b. 125.4 MHz
    c. 131.6 MHz
    d. 157.5 MHz

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Chapter 4: Frequency Modulation

1. The amount of frequency deviation from the carrier center frequency in an FM transmitter is proportional to what characteristic of the modulating signal?
    - **a. Amplitude**
    - b. Frequency
    - c. Phase
    - d. Shape

2. Both FM and PM are types of kind of modulation?
    - a. Amplitude
    - b. Phase
    - **c. Angle**
    - d. Duty Cycle

3. If the amplitude of the modulating signal decreases, the carrier deviation
    - a. Increases
    - **b. Decreases**
    - c. remains constant
    - d. goes to zero

4. On an FM signal, maximum deviation occurs at what point on the modulating signal?
    - a. Zero-signaling points
    - b. Peak positive peak amplitude
    - **c.** Peak negative peak amplitude
    - **d. Both b and c**

5. In PM, a frequency shift occurs while what characteristic of the modulating signal is changing?
    - a. Shape
    - b. Phase
    - c. Frequency
    - **d. Amplitude**

6. Maximum frequency deviation of a PM signal occurs at
    - **a. Zero crossing points**
    - b. Peak positive peak amplitude
    - c. Peak negative peak amplitude
    - d. Peak positive or negative peak amplitude

7. In PM, carrier frequency deviation is not proportional to:
    - a. Modulating signal amplitude
    - **b. Carrier amplitude and frequency**
    - c. Modulating signal frequency
    - d. Modulator phase shift

8. To compensate for increases in carrier frequency deviation with an increase in modulating signal frequency, what circuit is used between the modulating signal and the phase modulator?
    - **a. Low-pass filter**
    - b. High-pass filter
    - c. Phase shifter
    - d. Bandpass filter

9. The FM produced by PM is called
    - a. FM
    - b. PM
    - **c. Indirect FM**
    - d. Indirect PM

10. If the amplitude of the modulating signal applied to a phase modulator is constant, the output signal will be
    - a. Zero
    - **b. The carrier frequency**
    - c. Above the carrier frequency
    - d. Below the carrier frequency

11. A 100 MHz carrier is deviated 50 kHz by a 4 kHz signal. The modulation index is
    - a. 5
    - b. 8
    - **c. 12.5**
    - d. 20

12. The maximum deviation of an FM carrier is 2 kHz by a maximum modulating signal of 400 Hz. The deviation ratio is
    - a. 0.2
    - **b. 5**
    - c. 8
    - d. 40

13. A 70 kHz carrier has a frequency deviation of 4 kHz with a 1000 Hz signal. How many significant sideband pairs are produced?
    - a. 4
    - b. 5
    - c. 6
    - **d. 7**

14. What is the bandwidth of the FM signal described in question 13 above?
    - a. 4 kHz
    - b. 7 kHz
    - **c. 14 kHz**
    - d. 28 kHz

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Chapter 4: Frequency Modulation

15. What is the relative amplitude of the third pair of sidebands of an FM signal with m = 6
    - **a. 0.11**
    - b. 0.17
    - c. 0.24
    - d. 0.36

16. A 200 kHz carrier is modulated by a 2.5 kHz signal. The fourth pair of sidebands are spaced from the carrier by
    - a. 2.5 kHz
    - **b.** 5 kHz
    - **c. 10 kHz**
    - d. 15 kHz

17. An FM transmitter has a maximum deviation of 12 kHz and a maximum modulating frequency of 12 kHz. The bandwidth by Carson's rule is
    - a. 24 kHz
    - b. 33.6 kHz
    - c. 36.8 kHz
    - **d. 48 kHz**

18. The maximum allowed deviation of the FM sound signal in TV is 25 kHz. If the actual deviation is 18 kHz, the percent modulation is
    - a. 43%
    - **b. 72%**
    - c. 96%
    - d. 139%

19. Which of the following is not a major benefit of FM over AM?
    - a. Greater efficiency
    - b. Noise Immunity
    - c. Capture Effect
    - **d. Lower complexity and cost**

20. The primary disadvantage of FM is its
    - a. Higher cost and complexity
    - **b. Excessive use of spectrum space**
    - c. Noise susceptibility
    - d. Lower efficiency

21. Noise is primarily
    - **a. High frequency spikes**
    - b. Low-frequency variations
    - c. Random level shifts
    - d. Random frequency variations

22. The receiver circuit that rids FM of noise is the
    - a. Modulator
    - b. Demodulator
    - **c. Limiter**
    - d. Low-pass filter

23. The phenomenon of a strong FM signal dominating a weaker signa on a common frequency is referred to as the
    - **a. Capture effect**
    - b. Blot out
    - c. Quieting factor
    - d. Domination syndrome

24. The AM signals generated at a low level may only be amplified by what type of amplifier?
    - a. Op Amp
    - **b. Linear**
    - c. Class C
    - d. Push-pull

25. Frequency modulation transmitters are more efficient because their power is increased by what type of amplifier
    - a. Class A
    - b. Class B
    - **c. Class C**
    - d. All of the above

26. Noise interferes mainly with modulating signals that are
    - a. Sinusoidal
    - b. Nonsinusoidal
    - c. Low Frequency
    - **d. High frequencies**

27.Pre-emphasis circuits boost what modulating frequencies before modulation?
    - **a. High frequencies**
    - b. Mid-range frequencies
    - c. Low frequencies
    - d. All of the above

28. A pre-emphasis circuit is a
    - a. Low-pass filter
    - **b. High-pass filter**
    - c. Phase shifter
    - d. Bandpass filter

29. Pre-emphasis is compensated for at the receiver by

   a. Phase Inverter
   b. Bandpass filter
   c. High-pass filter
   d. Low-pass filter

30. The cut-off frequency of pre-emphasis and de-emphasis circuits is

   a. 1 kHz
   b. 2.122 kHz
   c. 5 kHz
   d. 75 kHz

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Chapter 5: Frequency Modulation Circuits

1. Another name for a VVC is
   a. PIN diode
   b. Varactor diode
   c. Snap diode
   d. Hot-carrier diode

2. The depletion region in a junction diode forms what part of a capacitor
   a. Plates
   b. Leads
   c. Package
   d. Dielectric

3. Increasing the reverse bias on a varactor diode will cause its capacitance
   a. Decrease
   b. Increase
   c. Remain the same
   d. Drop to zero

4. The capacitance of a varactor diode is in what general range
   a. pF
   b. nF
   c. μF
   d. F

5. In Fig. 5-3, the varactor diode is biased by which components?
   a. R_1, R_2
   b. R_1, C_2
   c. L_1, C_1
   d. RFC, C_3

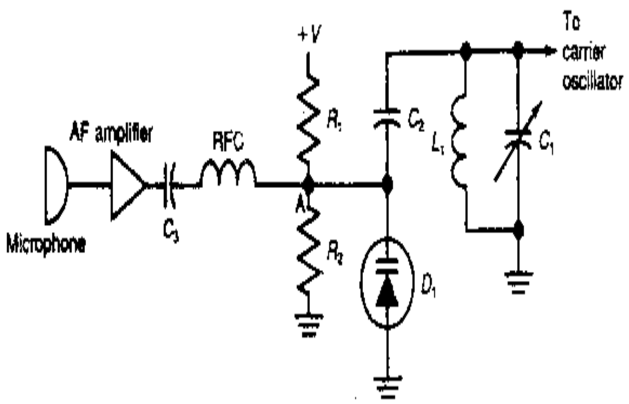

Fig. 5-3 Frequency modulation with a VVC.

6. In Fig. 5-3, if the reverse bias on D_1 is reduced, the resonant frequency of C_1
   a. Increase
   b. Decreases
   c. Remains the same
   d. Cannot be determined

7. The frequency change of a crystal oscillator produced by a varactor diode is
   a. Zero
   b. Small
   c. Medium
   d. Large

8. A phase modulator varies the phase shift of the
   a. Carrier
   b. Modulating Signal
   c. Both a and b
   d. Neither a nor b

9. The widest phase variation is obtained with a(n)
   a. RC low-pass filter
   b. RC high-pass filter
   c. LR low-pass filter
   d. LC resonant circuit

10. In Fig. 5-7, R_4 is the
   a. Pre-emphasis circuit
   b. De-emphasis circuit
   c. Deviation control
   d. Frequency determining component in the tuned circuit

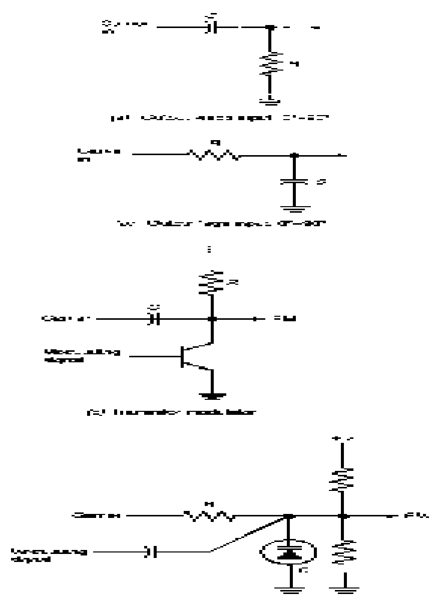

Fig. 5-7 Simple FM circuits.

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Chapter 5: Frequency Modulation Circuits

11. The small frequency change produced by a phase modulator can be increased by using a(n)
   a. Amplifier
   b. Mixer
   c. Frequency multiplier
   d. Frequency Divider

12. A crystal oscillator whose frequency can be changed by an input voltage is called a(n)
   a. VCO
   b. VXO
   c. VFO
   d. VHF

13. Which oscillators are preferred for carrier generators because of their good frequency stability?
   a. LC
   b. RC
   c. LR
   d. Crystal

14. Which of the following frequency demodulators requires an input limiter?
   a. Foster-Seeley discriminator
   b. Pulse-averaging discriminator
   c. Quadrature discriminator
   d. PLL

15. Which discriminator averages pulses in a low-pass filter?
   a. Radio detector
   b. PLL
   c. Quadrature detector
   d. Foster-Seeley discriminator

16. Which frequency demodulator is considered the best overall?
   a. Radio Detector
   b. PLL
   c. Quadrature
   d. Pulse-averaging discriminator

17. In Fig. 5-8, the voltage at point A when the input frequency is below the FM center frequency is
   a. Negative
   b. Positive
   c. Zero
   d. Indeterminant

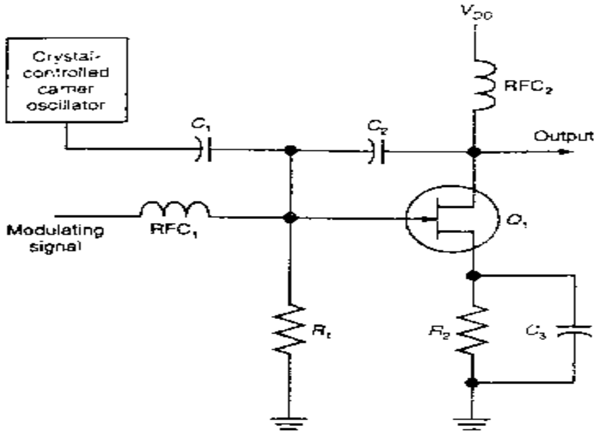

Fig. 5-8 An improved phase modulator.

18. In Fig. 5-8, R_3 and C_6 form which kind of circuit?
   a. Carrier Filter
   b. Pulse-averaging filter
   c. Pre-emphasis
   d. De-emphasis

19. In Fig. 5-10, the voltage across C_6 is
   a. Inversely proportional to signal amplitude
   b. Directly proportional to signal amplitude
   c. Directly proportional to frequency deviation
   d. Constant

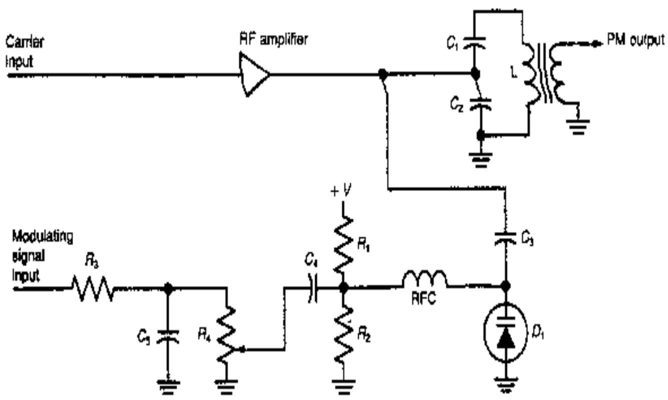

Fig. 5-10 One form of phase modulator.

20. In a pulse averaging discriminator, the pulses are produced by a(n)
   a. Astable multivibrator
   b. Zero-crossing detector
   c. One shot
   d. Low-pass filter

21. A reactance modulator looks like a capacitance of 35 pF in parallel with the oscillator-tuned circuit whose inductance is 50 μH and capacitance is 40 pF. What is the center frequency of the oscillator prior to FM?
    a. 1.43 MHz
    b. 2.6 MHz
    c. 3.56 MHz
    d. 3.8 MHz

22. Which of the following is true about the NE566 IC?
    a. It is a VCO
    b. Its output is sinusoidal
    c. It is an FM demodulator
    d. It uses LC-tuned circuits

23. An FM demodulator that uses a differential amplifier and tuned circuits to convert frequency variations into voltage variations is the
    a. Quadrature detector
    b. Foster-Seeley discriminator
    c. Differential peak detector
    d. Phase-Locked Loop

24. The output amplitude of the phase detector in a quadrature detector is proportional to
    a. Pulse width
    b. Pulse frequency
    c. Input amplitude
    d. The phase shift value at center frequency

25. The input to a PLL is 2 MHz. In order for the PLL to be locked, the VCO output must be
    a. 0 MHz
    b. 1 MHz
    c. 2 MHz
    d. 4 MHz

26. Decreasing the input frequency to a locked PLL will cause the VCO output to
    a. Decrease
    b. Increase
    c. Remain constant
    d. Jump to the free-running frequency

27. The range of frequencies over which a PLL will track input signal variations is known as the
    a. Circuit bandwidth
    b. Capture range
    c. Band of Acceptance
    d. Lock range

28. The band of frequencies over which a PLL will acquire or recognize an input signal is called the
    a. Circuit Bandwidth
    b. Capture range
    c. Band of acceptance
    d. Lock range

29. Over a narrow range of frequencies, the PLL acts like a
    a. Low-pass filter
    b. Bandpass filter
    c. Tunable oscillator
    d. Frequency modulator

30. The output of a PLL frequency demodulator is taken from
    a. Low-pass filter
    b. VCO
    c. Phase detector
    d. None of the above

COMMUNICATION ELECTRONICS
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Chapter 6: Radio Transmitters

1. Which of the following circuits is not typically part of every radio transmitter?
   a. Driver Amplifier
   b. Carrier Oscillator
   c. Mixer
   d. Final Power Amplifier

2. Class C amplifiers are not used in which type of transmitter?
   a. AM
   b. SSB
   c. CW
   d. FM

3. A circuit that isolates the carrier oscillator from load changes is called a
   a. Final Amplifier
   b. Driver Amplifier
   c. Linear Amplifier
   d. Buffer Amplifier

4. A class B amplifier conducts for how many degrees of an input sine wave?
   a. 90° - 150°
   b. 180°
   c. 180° - 360°
   d. 360°

5. Bias for a class C amplifier produced by an input RC network is known as
   a. Signal Bias
   b. Self Bias
   c. Fixed External Bias
   d. Threshold Bias

6. An FM transmitter has a 9 MHz crystal carrier oscillator and frequency multipliers of 2, 3, 4. The output frequency is
   a. 54 MHz
   b. 108 MHz
   c. 216 MHz
   d. 288 MHz

7. The most efficient RF power amplifier is which class amplifier?
   a. Class B
   b. Class A
   c. Class AB
   d. Class C

8. Collector current in a class C amplifier is a
   a. Square Wave
   b. Sine Wave
   c. Pulse
   d. Half Sine Wave

9. The maximum power of typical transistor RF power amplifiers is in what range?
   a. Kilowatts
   b. Milliwatts
   c. Hundreds of Watts
   d. Watts

10. Self-oscillation in a transistor amplifier is usually caused by
   a. Excessive Gain
   b. Stray Inductance
   c. Internal Capacitance
   d. Unmatched Impedances

11. Neutralization is the process of
   a. Cancelling the effect of internal device capacitance
   b. Bypassing undesired alternating current
   c. Reducing gain
   d. Eliminating Harmonics

12. Maximum power transfer occurs when what relationship exists between the generator impedance Z_I and the load impedance Z_l?
   a. $Z_i = Z_l$
   b. $Z_i > Z_l$
   c. $Z_i < Z_l$
   d. $Z_i = 0\Omega$

13. Which of the following is not a benefit of a toroid RF inductor?
   a. Higher Q
   b. No shielding required
   c. Fewer turns of wire
   d. Self - supporting

14. A toroid is a
   a. Coil Holder
   b. Type of Inductor
   c. Magnetic Core
   d. Transformer

COMMUNICATION ELECTRONICS
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Chapter 6: Radio Transmitters

15. Which of the following is not commonly used for impedance matching in a transmitter?
    - **a. Resistive attenuator**
    - b. Transformer
    - c. L Network
    - d. T Network

16. To a match a 6 Ω amplifier impedance to a 72 Ω antenna load, a transformer must have a turns ration N_P/N_S of
    - a. 0.083
    - **b. 0.289**
    - c. 3.46
    - d. 12

17. Impedance matching in a broadband linear RF amplifier is handled with a(n)
    - a. Pi Network
    - b. L Network
    - c. Parallel Tuned Circuit
    - **d. Balun**

18. A class C amplifier has a supply voltage of 24 V and a collector current of 2.5 A. Its efficiency is 80 %. The RF output power is
    - a. 24 W
    - **b. 48 W**
    - c. 60 W
    - d. 75 W

19. Which of the following is not a benefit of speech-processing circuits?
    - **a. Improved frequency stability**
    - b. Increased average output power
    - c. Limited bandwidth
    - d. Prevention of overmodulation

20. In an AM transmitter, a clipper circuit eliminates
    - a. Harmonics
    - **b. Splatter**
    - c. Overdeviation
    - d. Excessive gain

21. In a speech-processing circuit, a low-pass filter prevents
    - a. Overdeviation
    - b. Overmodulation
    - c. High gain
    - **d. Excessive signal bandwidth**

22. The gain of a transistor amplifier is
    - a. Inversely proportional to collector current
    - b. Directly proportional to frequency
    - **c. Directly proportional to collector current**
    - d. Inversely proportional to frequency

23. What values of L and C in an L network are required to match a 10 Ω transistor amplifier impedance to a 50 Ω load at 27 MHz?
    - a. L = 47 nH, C = 185 pF
    - **b. L = 118 nH, C = 236 pF**
    - c. L = 0.13 μH, C = 220 pF
    - d. L = 0.3 μH, C = 330 pF

COMMUNICATION ELECTRONICS
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Chapter 7: Communications Receivers

1. The simplest receiver is a(n)
    a. RF amplifier
    b. Demodulator
    c. AF amplifier
    d. Tuned circuit

2. The key conceptual circuit in a superhet receiver is the
    a. Mixer
    b. RF amplifier
    c. Demodulator
    d. AF amplifer

3. Most of the gain and selectivity in a superhet is obtained in the
    a. RF amplifier
    b. Mixer
    c. IF amplifier
    d. AF amplifier

4. The sensitivity of a receiver depends upon the receiver's overall
    a. Bandwidth
    b. Selectivity
    c. Noise response
    d. Gain

5. The ability of a receiver to separate one signal from others on closely adjacent frequencies is called the
    a. Sensitivity
    b. S/N ratio
    c. Selectivity
    d. Gain

6. A mixer has a signal input of 50 MHz and an LO frequency of 59 MHz. The IF is
    a. 9 MHz
    b. 50 MHz
    c. 59 MHz
    d. 109 MHz

7. A signal 2 times the IF away from the desired signal that causes interference is referred to as a(n)
    a. Ghost
    b. Image
    c. Phantom
    d. Inverted Signal

8. A receiver has a desired input signal of 18 MHz and an LO frequency of 19.6 MHz. The image frequency is
    a. 1.6 MHz
    b. 18 MHz
    c. 19.6 MHz
    d. 21.2 MHz

9. The main cause of image interference is
    a. Poor front-end selectivity
    b. Low gain
    c. A high IF
    d. A low S/N ratio

10. For best image rejection, the IF for a 30 MHz signal would be
    a. 455 kHz
    b. 3.3 MHz
    c. 9 MHz
    d. 55 MHz

11. A tuned circuit is resonant at 4 MHz. Its Q is 100. The bandwidth is
    a. 400 Hz
    b. 4 kHz
    c. 40 kHz
    d. 400 kHz

12. A crystal filter has a 6 dB bandwidth of 2.6 kHz and a 60 dB bandwidth of 14 kHz. The shape factor is
    a. 0.186
    b. 5.38
    c. 8.3
    d. 36.4

13. Most internal noise comes from
    a. Shot noise
    b. Transit-time noise
    c. Thermal agitation
    d. Skin effect

14. Which of the following is not a source of external noise
    a. Thermal agitation
    b. Auto ignitions
    c. The sun
    d. Fluorescent lights

COMMUNICATION ELECTRONICS
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Chapter 7: Communications Receivers

15. Noise can be reduced by
    a. Widening the bandwidth
    b. Narrowing the bandwidth
    c. Increasing temperature
    d. Increasing transistor current levels

16. Noise at the input to a receiver can be as high as several
    a. Microvolts
    b. Millivolts
    c. Volts
    d. Kilovolts

17. Which circuit contributes most to the noise in a receiver?
    a. IF amplifier
    b. Demodulator
    c. AF amplifier
    d. Mixer

18. Which noise figure represents the lowest noise
    a. 1.6 dB
    b. 2.1 dB
    c. 2.7 dB
    d. 3.4 dB

19. Which filter shape factor represents the best skirt selectivity?
    a. 1.6
    b. 2.1
    c. 5.3
    d. 8

20. Which input signal below represents the best receiver sensitivity?
    a. 0.5 μV
    b. 1 μV
    c. 1.8 μV
    d. 2 μV

21. Transistor with the lowest noise figure in the microwave region is a(n)
    a. MOSFET
    b. Dual-gate MOSFET
    c. JFET
    d. MESFET

22. The AGC circuits usually control the gain of the
    a. Mixer
    b. Detector
    c. IF amplifiers
    d. Audio amplifiers

23. Selectivity is obtained in most receivers from
    a. Crystal filters
    b. Mechanical filters
    c. Double-tuned circuits
    d. Audio filters

24. Widest bandwidth in a double-tuned circuit is obtained with
    a. Undercoupling
    b. Critical coupling
    c. Optimum coupling
    d. Overcoupling

25. Automatic gain control permits a wide range of signal amplitudes to be accommodated by controlling the gain of the
    a. RF amplifier
    b. IF amplifier
    c. Mixer
    d. AF amplifier

26. In an IF amplifier with reverse AGC, a strong signal will cause the collector current to
    a. Increase
    b. Decrease
    c. Remain the same
    d. Drop to zero

27. Usually AGC voltage is derived by the
    a. RF amplifier
    b. IF amplifier
    c. Demodulator
    d. AF amplifier

28. An AFC circuit is used to correct for
    a. Audio distortion
    b. Strong input signals
    c. Instability in the IF amplifier
    d. Frequency drift in the LO

29. A circuit keeps the audio cut off until a signal is received is known as
    a. A squelch
    b. AFC
    c. AGC
    d. A noise blanker

30. A BFO is used in the demodulation of which types of signals?
    a. AM
    b. FM
    c. SSB or CW
    d. QPSK

31. Which of the following circuits are not typically shared in an SSB transceiver?
    a. Crystal filter
    b. Mixers
    c. Power supply
    d. LO

32. The basic frequency synthesizer circuit is a(n)
    a. Mixer
    b. Frequency multiplier
    c. Frequency divider
    d. PLL

33. The output frequency increment of a frequency synthesizer is determined by the
    a. Frequency division ratio
    b. Reference input to the phase detector
    c. Percentage of output frequency
    d. Frequency multiplication factor

34. The output of the frequency synthesizer is changed by varying the
    a. Reference frequency input to the phase detector
    b. Frequency division ratio
    c. Frequency multiplication factor
    d. Mixer LO frequency

35. In Fig 7-28, if the input reference is 25 kHz and the divide ratio is 144, the VCO output frequency
    a. 173.61 Hz
    b. 144 kHz
    c. 3.6 MHz
    d. 5.76 MHz

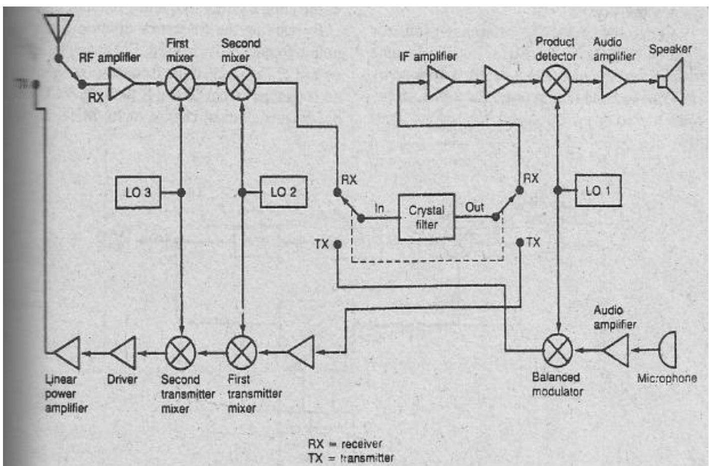

Fig. 7-28 An SSB transceiver showing circuit sharing.

36. The bandwidth of a parallel LC circuit can be increased by
    a. Increasing X_C
    b. Decreasing X_L
    c. Decreasing coil resistance
    d. A resistor connected in parallel

37.The upper and lower cutoff frequencies of a tuned circuit are 1.7 and 1.5 MHz respectively. The circuit Q is
    a. 8
    b. 10
    c. 16
    d. 24

38. The noise voltage across a 300 Ω input resistance to a TV set with a 6 MHz bandwidth and a temperature of 30°C is
    a. 2.3 μV
    b. 3.8 μV
    c. 5.5 μV
    d. 6.4 μV

39. The stage gains in a superheterodyne are follows RF amplifier, 10dB; mixer, 6dB; two IF amplifiers, each 33 dB; detector, -4 dB; AF amplifier, 28 dB. The total gain is
    a. 73 dB
    b. 82 dB
    c. 106 dB
    d. 139 dB

40. A tuned circuit resonates at 12 MHz with an inductance of 5 μH whose resistance is 6 Ω. The circuit bandwidth is
    a. 98 kHz
    b. 191 kHz
    c. 754 kHz
    d. 1.91 MHz

41. In a receiver with noise-derived squelch, the presence of an audio signal causes the audio amplifier to be
    a. Enabled
    b. Disabled

COMMUNICATION ELECTRONICS
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Chapter 8: Multiplexing

1. Multiplexing is the process of
   a. Several signal sources transmitting simultaneously to a receiver on a common frequency
   b. Sending the same signal over multiple channels to multiple destinations
   c. Transmitting multiple signals over multiple channels
   d. Sending multiple signals simultaneously over a single channel

2. In FDM, multiple signals
   a. Transmit at different times
   b. Share a common bandwidth
   c. Use multiple channels
   d. Modulate one another

3. Each signal in a an FDM system
   a. Modulates a subcarrier
   b. Modulates the final carrier
   c. Is mixed with all the others before modulation
   d. Serves as a subcarrier

4. Frequency modulation in FDM systems is usually accomplished with a
   a. Reactance modulator
   b. Varactor
   c. VCO
   d. PLL

5. Which of the following is not a typical FDM application
   a. Telemetry
   b. Stereo Broadcasting
   c. Telephone
   d. Secure communications

6. The circuit the performs demultiplexing in an FDM system is a(n)
   a. Op amp
   b. Bandpass filter
   c. Discriminator
   d. Subcarrier oscillator

7. Most FDM telemetry systems use
   a. AM
   b. FM
   c. SSB
   d. PSK

8. The best frequency demodulator is the
   a. PLL discriminator
   b. Pulse-averaging discriminator
   c. Foster-Seeley discriminator
   d. Ratio detector

9. The modulation used in FDM telephone systems is
   a. AM
   b. FM
   c. SSB
   d. PSK

10. The FDM telephone systems accommodate many channels by
   a. Increasing the multiplexer size
   b. Using many final carriers
   c. Narrowing the bandwidth of each
   d. Using multiple levels of multiplexing

11. In FM stereo broadcasting, the L + R signal
   a. Double-sideband modulates a subcarrier
   b. Modulates the FM carrier
   c. Frequency modulates a subcarrier
   d. Is not transmitted

12. In FM stereo broadcasting, the L – R signal
   a. Double-sideband modulates a subcarrier
   b. Modulates the FM carrier
   c. Frequency modulates a subcarrier
   d. Is not transmitted

13. The SCA signal if used in FM broadcasting is transmitted via
   a. A 19-kHZ subcarrier
   b. A 38-kHz subcarrier
   c. A 67-kHz subcarrier
   d. The main FM carrier

14. In TDM, multiple signals
   a. Share a common bandwidth
   b. Modulate subcarriers
   c. Are sampled at high speeds
   d. Take turns transmitting

15. In TDM, each signal may use he full bandwidth of the channel
   a. True
   b. False

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Chapter 8: Multiplexing

16. Sampling an analog signal produces
    - **a. PAM**
    - b. AM
    - c. FM
    - d. PCM

17. The maximum bandwidth that an analog signal use with a sampling frequency of 108 kHz is
    - a. 27 kHz
    - **b. 54 kHz**
    - c. 108 kHz
    - d. 216 kHz

18. Pulse-amplitude modulation signals are multiplexed by using
    - a. Subcarriers
    - b. Bandpass filters
    - c. A/D converters
    - **d. FET switches**

19. In PAM demultiplexing, the receiver clock is derived from
    - a. Standard radio station WWV
    - b. A highly accurate internal oscillator
    - **c. The PAM signal itself**
    - d. The 60-Hz power line

20. In PAM/TDM system, keeping the multiplexer and DEMUX channels step with one another is done by a
    - a. Clock recovery circuit
    - **b. Sync pulse**
    - c. Sampling
    - d. Sequencer

21. Transmitting data as serial binary words is called
    - a. Digital communications
    - b. Quantizing
    - c. PAM
    - **d. PCM**

22. Converting analog signals to digital is done by sampling and
    - **a. Quantizing**
    - b. Companding
    - c. Pre-emphasis
    - d. Mixing

23. A quantizer is a(n)
    - a. Multiplexer
    - b. Demultiplexer
    - **c. A/D converter**
    - d. D/A converter

24. Emphasizing low-level signals and compressing higher-level signals is called
    - a. Quantizing
    - **b. Companding**
    - c. Pre-emphasis
    - d. Sampling

25. Which of the following is not a benefit of companding?
    - a. Minimizes noise
    - b. Minimizes number of bits
    - c. Minimizes quantizing error
    - **d. Minimizes signal bandwidth**

26. A telephone system using TDM and PCM is called
    - a. PBX
    - b. RS-232
    - **c. T-1**
    - d. Bell 212

27. An IC that contains A/D and D/A converters, companders and parallel-to-serial converters is called
    - **a. Codec**
    - b. Data converter
    - c. Multiplexer
    - d. Modem

28. Pulse-code modulation is preferred to PAM because of its
    - a. Resistance to quantizing error
    - b. Simplicity
    - c. Lower cost
    - **d. Superior noise immunity**

COMMUNICATION ELECTRONICS
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Chapter 9: Antennas and Transmission Lines

1. The most commonly used transmission line is a
    a. two-wire balanced line
    b. single wire
    c. three-wire line
    d. coax

2. The characteristic impedance of a transmission line does not depend upon its
    a. length
    b. conductor diameter
    c. conductor spacing
    d. none of the above

3. Which of the following is not a common transmission line impedance?
    a. 50 Ω
    b. 75 Ω
    c. 120 Ω
    d. 300 Ω

4. For maximum absorption of power at the antenna, the relationship between the characteristic impedance of the line Z_O and the load impedance Z_l should be
    a. $Z_O = Z_l$
    b. $Z_O > Z_l$
    c. $Z_O < Z_l$
    d. $Z_O = 0$

5. The mismatch between antenna and transmission line impedances cannot be corrected for by
    a. using an LC matching network
    b. adjusting antenna length
    c. using a balun
    d. adjusting the length of transmission line

6. A pattern of voltage and current variations along a transmission line not terminated in its characteristic impedance is called
    a. an electric field
    b. radio waves
    c. standing waves
    d. a magnetic field

7. The desirable SWR on a transmission line is
    a. 0
    b. 1
    c. 2
    d. infinity

8. A 50 Ω coax is connected to a 73 Ω antenna. The SWR is
    a. 0.685
    b. 1
    c. 1.46
    d. 2.92

9. The most desirable reflection coefficient is
    a. 0
    b. 0.5
    c. 1
    d. infinity

10. A ratio expressing the percentage of incident voltage reflected on a transmission line is know as the
    a. velocity factor
    b. standing wave ratio
    c. reflection coefficient
    d. line efficiency

11. The minimum voltage along a transmission line is 260 V, while the maximum voltage is 390 V. The SWR is
    a. 0.67
    b. 1.0
    c. 1.2
    d. 1.5

12. Three feet is one wavelength at a frequency of
    a. 100 MHz
    b. 164 MHz
    c. 300 MHz
    d. 328 MHz

13. At very high frequencies, transmission lines are used as
    a. tuned circuits
    b. antennas
    c. insulators
    d. resistors

COMMUNICATION ELECTRONICS
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Chapter 9: Antennas and Transmission Lines

14. A shorted quarter-wave line at the operating frequency acts like a(n)
   a. series resonant circuit
   b. parallel resonant circuit
   c. capacitor
   d. inductor

15. A shorted half-wave line at the operating frequency acts like a(n)
   a. capacitor
   b. inductor
   c. series resonant circuit
   d. parallel resonant circuit

16. A popular half-wavelength antenna is the
   a. ground plane
   b. end-fire
   c. collinear
   d. dipole

17. The length of a doublet at 27 MHz is
   a. 8.67 ft
   b. 17.3 ft
   c. 18.2 ft
   d. 34.67 ft

18. A popular vertical antenna is the
   a. collinear
   b. dipole
   c. ground plane
   d. broadside

19. The magnetic field of an antenna is perpendicular to the earth. The antenna's polarization
   a. is vertical
   b. is horizontal
   c. is circular
   d. cannot be determined from the information given

20. An antenna that transmits or receives equally well in all directions is said to be
   a. omnidirectional
   b. bidirectional
   c. unidirectional
   d. quasidirectional

21. The horizontal radiation pattern of a dipole is a
   a. circle
   b. figure eight
   c. clover leaf
   d. narrow beam

22. The length of a ground plane vertical at 146 MHz is
   a. 1.6 ft
   b. 1.68 ft
   c. 2.05 ft
   d. 3.37 ft

23. The impedance of a dipole is about
   a. 50 Ω
   b. 73 Ω
   c. 93 Ω
   d. 300 Ω

24. A direction antenna with two or more elements is known as a(n)
   a. folded dipole
   b. ground plane
   c. loop
   d. array

25. The horizontal radiation pattern o a vertical dipole is
   a. figure eight
   b. circle
   c. narrow beam
   d. clover leaf

26. In a Yagi antenna, maximum direction of radiation is toward the
   a. director
   b. driven element
   c. reflector
   d. sky

27. Conductors in multielement antennas that do not receive energy directly from the transmission line are known as
   a. parasitic elements
   b. driven elements
   c. the boom
   d. receptors

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Chapter 9: Antennas and Transmission Lines

28. A coax has an attenuation of 2.4 db per 100 ft. The attenuation for 275 ft is
    a. 2.4 dB
    b. 3.3 dB
    c. 4.8 dB
    d. 6.6 dB

29. An antenna has a power gain of 15. The power applied to the antenna is 32 W. The effective radiated power is
    a. 15 W
    b. 32 W
    c. 120 W
    d. 480 W

30. Which beamwidth represents the best antenna directivity
    a. 7°
    b. 12°
    c. 19°
    d. 28°

31. The radiation pattern of collinear and broadside antennas
    a. omnidirectional
    b. bidirectional
    c. unidirectional
    d. clover-leaf shaped

32. Which antenna has a unidirectional radiation pattern and gain
    a. dipole
    b. ground plane
    c. yagi
    d. collinear

33. A wide-bandwidth multielement driven array is the
    a. end-fire
    b. log-periodic
    c. yagi
    d. collinear

34. Ground-wave communications is most effective in what frequency range?
    a. 300 kHz to 3 MHz
    b. 3 to 30 MHz
    c. 30 to 300 MHz
    d. above 300 MHz

35. The ionosphere causes radio signals to be
    a. diffused
    b. absorbed
    c. refracted
    d. reflected

36. The ionosphere has its greatest effect on signals in what frequency range?
    a. 300 kHz to 3 MHz
    b. 3 to 30 MHz
    c. 30 to 300 MHz
    d. above 300 MHz

37. The type of radio wave responsible for long-distance communications by multiple skips is the
    a. ground wave
    b. direct wave
    c. surface wave
    d. sky wave

38. Microwave signals propagate by way of the
    a. direct wave
    b. sky wave
    c. surface wave
    d. standing wave

39. The line-of-sight communications is not a factor in which frequency range?
    a. VHF
    b. UHF
    c. HF
    d. microwave

40. A microwave-transmitting antenna is 550 ft high. The receiving antenna is 200 ft high. The maximum transmission distance is
    a. 20 mi
    b. 33.2 mi
    c. 38.7 mi
    d. 53.2 mi

41. To increase the transmission distance of a UHF signal, which of the following should be done?
    a. increase antenna gain
    b. increase antenna height
    c. increase transmitter power
    d. increase receiver sensitivity

COMMUNICATION ELECTRONICS
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Chapter 9: Antennas and Transmission Lines

42. A coax has a velocity factor of 0.68. What is the length of a half wave at 30 MHz?
   - **a. 11.2 ft**
   - b. 12.9 ft
   - c. 15.6 ft
   - d. 16.4 ft

43. Which transmission line has the lowest attenuation?
   - **a. twin lead**
   - b. RG-11/U
   - c. RG-59/U
   - d. RG-214/U

44. Refer to Fig. 9-39. The beam width of this antenna patter is approximately
   - a. 30°
   - b. 38°
   - c. 45°
   - **d. 60°**

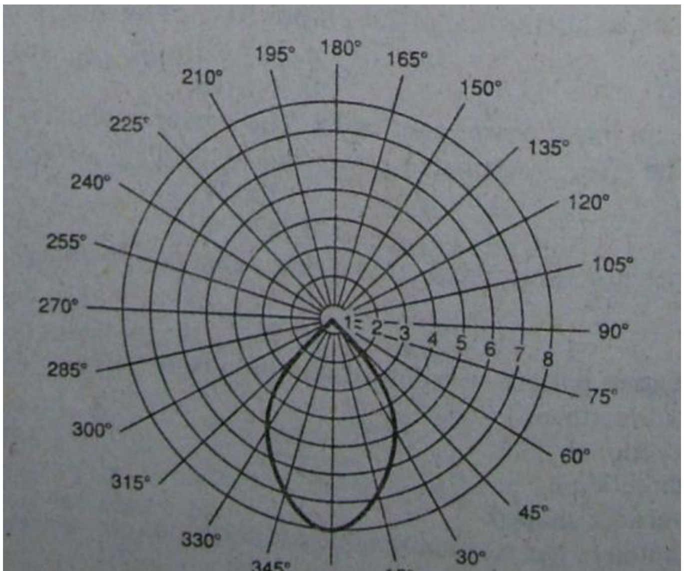

Fig. 9-39 Pattern for Chapter Review Question 9-44.

45. A receiver-transmitter station used to increase the communications range of VHF, UHF, and microwave signals is called a(n)
   - a. transceiver
   - b. remitter
   - **c. repeater**
   - d. amplifier

COMMUNICATION ELECTRONICS
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Chapter 10: Microwave Techniques

1. The main benefit of using microwaves is
   a. lower-cost equipment
   b. simpler equipment
   c. greater transmission distances
   d. more spectrum space

2. Radio communications are regulated in the United States by the
   a. Federal Trade Commission
   b. Congress
   c. Federal Communications Commission
   d. Military

3. Which of the following is not a disadvantage of microwaves?
   a. higher-cost equipment
   b. line-of-sight transmission
   c. conventional components are not usable
   d. circuits are more difficult to analyze

4. Which of the following is a microwave frequency
   a. 1.7 MHz
   b. 750 MHz
   c. 0.98 GHz
   d. 22 GHz

5. Which of the following is not a common microwave application?
   a. radar
   b. mobile radio
   c. telephone
   d. spacecraft communications

6. Coaxial cable is not widely used for long microwave transmission lines because of its
   a. high loss
   b. high cost
   c. large size
   d. excessive radiation

7. Stripline and microstrip transmission lines are usually made with
   a. coax
   b. parallel wires
   c. twisted pair
   d. PCBs

8. The most common cross section of a wave guide is a
   a. square
   b. circle
   c. triangle
   d. rectangle

9. A rectangular waveguide has a width of 1 in. and a height of 0.6 in. Its cutoff frequency is
   a. 2.54 GHz
   b. 3.0 GHz
   c. 5.9 GHz
   d. 11.8 GHz

10. A waveguide has a cutoff frequency of 17 GHz. Which of the signals will not be passed by the waveguide?
   a. 15 GHz
   b. 18 GHz
   c. 22 GHz
   d. 25 GHz

11. Signal propagation in a waveguide is by
   a. electrons
   b. electric and magnetic fields
   c. holes
   d. air pressure

12. When the electric field in a waveguide is perpendicular to the direction of wave propagation, the mode is said to be
   a. vertical polarization
   b. horizontal polarization
   c. transverse electric
   d. transverse magnetic

13. The dominant mode in most waveguides is
   a. $TE_{0,1}$
   b. $TE_{1,2}$
   c. $TM_{0,1}$
   d. $TM_{1,1}$

14. A magnetic field is introduced into a waveguide by a
   a. probe
   b. dipole
   c. stripline
   d. capacitor

15. A half-wavelength, closed section of a waveguide that acts as a parallel resonant circuit is known as a(n)
    a. half-wave section
    b. cavity resonator
    c. LCR circuit
    d. directional coupler

16. Decreasing the volume of a cavity causes its resonant frequency to
    a. increase
    b. decrease
    c. remain the same
    d. drop to zero

17. A popular microwave mixer diode is the
    a. Gunn
    b. Varactor
    c. Hot carrier
    d. IMPATT

18. Varactor and step-recovery diodes are widely used in what type of circuit
    a. amplifier
    b. oscillator
    c. frequency multiplier
    d. mixer

19. Which diode is a popular microwave oscillator
    a. IMPATT
    b. Gunn
    c. Varactor
    d. Mixer

20. Which type of diode does not ordinarily operate with reverse bias
    a. Varactor
    b. IMPATT
    c. Snap-off
    d. Tunnel

21. Low-power Gunn diodes are replacing
    a. Reflex klystrons
    b. TWTs
    c. Magnetrons
    d. Varactor diodes

22. Which of the following is not a microwave tube?
    a. travelling-wave tube
    b. cathode-ray tube
    c. klystron
    d. magnetron

23. In a klystron amplifier, velocity modulation of the electron beam is produced by the
    a. collector
    b. catcher cavity
    c. cathode
    d. buncher cavity

24. A reflex klystron is used as a(n)
    a. amplifier
    b. oscillator
    c. mixer
    d. frequency multiplier

25. For proper operation, a magnetron must be accompanied by a
    a. cavity resonator
    b. strong electric field
    c. permanent magnet
    d. high dc voltage

26. The operating frequency of klystrons and magnetrons is set by the
    a. cavity resonators
    b. DC supply voltage
    c. input signal frequency
    d. number of cavities

27. A magnetron is used only as a(n)
    a. amplifier
    b. oscillator
    c. mixer
    d. frequency multiplier

28. A common application for magnetrons is in
    a. radar
    b. satellites
    c. two-way radio
    d. TV sets

29. In a TWT, the electron beam is density-modulated by a
    a. permanent magnet
    b. modulation transformer
    c. helix
    d. cavity resonator

30. The main advantage of a TWT over a klystron for microwave amplification is
    a. lower cost
    b. smaller size
    c. higher power
    d. wider bandwidth

31. High-power TWTs are replacing what in microwave amplifiers?
    a. MESFETs
    b. Magnetrons
    c. Klystrons
    d. IMPATT diodes

32. The most widely used microwave antenna is a
    a. half-wave dipole
    b. quarter-wave probe
    c. single loop
    d. horn

33. What happens when a horn antenna is made longer?
    a. gain increases
    b. beam width increases
    c. both a and b
    d. neither a nor b

34. A pyramidal horn used at 5 GHz has an aperture that is 7 by 9 cm. The gain is about
    a. 10.5 dB
    b. 11.1 dB
    c. 22.6 dB
    d. 35.8 dB

35. Given the frequency and dimensions in Question 34 above the beamwidth is about
    a. 27°
    b. 53°
    c. 60°
    d. 80°

36. The diameter of a parabolic reflector should be at least how many wavelengths at the operating frequency?
    a. 1
    b. 2
    c. 5
    d. 10

37. The point where the antenna is mounted with respect to the parabolic reflector is called
    a. focal point
    b. center
    c. locus
    d. tangent

38. Using a small reflector to beam waves to the larger parabolic reflector is known as
    a. focal feed
    b. horn feed
    c. cassegrain feed
    d. coax feed

39. Increasing the diameter of a parabolic reflector causes which of the following
    a. decrease beamwidth
    b. increase gain
    c. increase beam width
    d. a and b
    e. b and c
    f. none of the above

40. A helical antenna is made up of a coil and a
    a. director
    b. reflector
    c. dipole
    d. horn

41. The output of a helical antenna is
    a. vertically polarized
    b. horizontally polarized
    c. circularly polarized
    d. both a and b

42. A common omnidirectional microwave antenna is the
    a. horn
    b. parabolic reflector
    c. helical
    d. bicone

COMMUNICATION ELECTRONICS
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Chapter 11: Introduction to Satellite Communications

1. As the height of a satellite orbit gets lower, the speed of the satellite
    - **a. Increases**
    - b. Decreases
    - c. Remains the same
    - d. None of the above

2. The main functions of a communications satellite is a a(a)
    - **a. Repeater**
    - b. Reflector
    - c. Beacon
    - d. Observation platform

3. The key electronic component in a communications satellite is the
    - a. Telemetry
    - b. On board computer
    - c. Command and control system
    - **d. Transponder**

4. A circular orbit around the equator with a 24 h period is called a(n)
    - a. Elliptical orbit
    - **b. Geostationary orbit**
    - c. Polar orbit
    - d. Transfer orbit

5. A satellite stays in orbit because the following 2 factor are balanced
    - a. Satellite weight and speed
    - **b. Gravitational pull and inertia**
    - c. Centripetal force and speed
    - d. Satellite weight and the pull of the moon and sun

6. The height of a satellite in a synchronous equatorial orbit is
    - a. 100 mi
    - b. 6800 mi
    - **c. 22,300 mi**
    - d. 35,860 mi

7. Most satellites operate in which frequency band?
    - a. 30 to 300 MHz
    - b. 300 MHz to 3GHz
    - **c. 3 GHz to 30 GHz**
    - d. above 300 GHz

8. The main power sources for a satellite are
    - a. Batteries
    - **b. Solar cells**
    - c. Fuel cells
    - d. Thermoelectric generators

9. The maximum height of an elliptical orbit is called
    - a. Perigee
    - b. Apex
    - c. Zenith
    - **d. Apogee**

10. Batteries are used to power all satellite subsystems
    - a. At all times
    - b. Only during emergencies
    - **c. During ellipse periods**
    - d. To give the solar arrays a rest

11. The satellite subsystem that monitors and controls the satellite is the
    - a. Propulsion subsystem
    - b. Power subsystem
    - c. Communications subsystems
    - **d. Telemetry, tracking and command system**

12. The basic technique used to stabilize a satellite is
    - a. Gravity-forward motion balance
    - **b. Spin**
    - c. Thruster control
    - d. Solar panel orientation

13. The jet thrusters are usually fired to
    - **a. Maintain altitude**
    - b. Put the satellite into the transfer orbit
    - c. Inject the satellite into the geosynchronous orbit
    - d. Bring the satellite back to the earth

14. Most commercial satellite activity occurs in which bands?
    - a. L
    - **b. C and Ku**
    - c. X
    - d. S and P

COMMUNICATION ELECTRONICS
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Chapter 11: Introduction to Satellite Communications

15. How can multiple earth stations share a satellite on the same frequency
    - **a. Frequency reuse**
    - b. Multiplexing
    - c. Mixing
    - d. They can't

16. The typical bandwidth of a satellite band is
    - a. 36 MHz
    - b. 40 MHz
    - c. 70 MHz
    - **d. 500 MHz**

17. Which of the following is not usually a part of a transponder
    - a. LNA
    - b. Mixer
    - **c. Modulator**
    - d. HPA

18. The satellite communications channels in a transponder are defined by the
    - a. LNA
    - **b. Bandpass filter**
    - c. Mixer
    - d. Input signals

19. The HPAs in most satellites are
    - a. TSTs
    - **b. Klystrons**
    - c. Vacuum tubes
    - d. Magnetrons

20. The physical location of a satellite is determined by its
    - a. Distance from the earth
    - **b. Latitude and longitude**
    - c. Reference to the stars
    - d. Position relative to the sun

21. The receive GCE system in an earth station performs what function(s)
    - a. Modulation and multiplexing
    - b. Up conversion
    - **c. Demodulation and demultiplexing**
    - d. Down conversion

22. Which of the following types of HPA is not used in earth stations
    - a. TWT
    - b. Transistor
    - c. Klystron
    - **d. Magnetron**

23. A common up-converter and down-converter IF is
    - a. 36 MHz
    - b. 40 MHz
    - **c. 70 MHz**
    - d. 500 MHz

24. The type of modulation used on voice and video signals is
    - a. AM
    - **b. FM**
    - c. SSB
    - d. QPSK

25. The modulation normally used with digital data is
    - a. AM
    - b. FM
    - c. SSB
    - **d. QPSK**

26. Which of the following is not a typical output from a GPS receiver?
    - a. Latitude
    - **b. Speed**
    - c. Altitude
    - d. Longitude

COMMUNICATION ELECTRONICS
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Chapter 12: Data Communications

1. Data communications refer to the transmission of
   a. Voice
   b. Video
   c. Computer data
   d. All of the above

2. Data communications uses
   a. Analog methods
   b. Digital methods
   c. Either of the above
   d. Neither of the above

3. Which of the following is not primarily a type of data communications
   a. Telephone
   b. Teletype
   c. Telegraph
   d. CW

4. The main reason that serial transmission is preferred to parallel transmission is that
   a. Serial is faster
   b. Serial requires only a single channel
   c. Serial requires multiple channels
   d. Parallel is too expensive

5. Mark and space refer respectively to
   a. Dot and dash
   b. Messege and interval
   c. Binary 1 and binary 0
   d. On and off

6. The number of amplitude, frequency, or phase changes that take place per second is known as the
   a. Data rate in bits per second
   b. Frequency of operation
   c. Speed limit
   d. Baud rate

7. Data transmission of one character at a time with start and stop bits is known as what type of transmission?
   a. Asynchronous
   b. Serial
   c. Synchronous
   d. Parallel

8. The most widely used data communications code is
   a. Morse
   b. ASCII
   c. Baudot
   d. EBCDIC

9. The ASCII code has
   a. 4 bits
   b. 5 bits
   c. 7 bits
   d. 8 bits

10. Digital signals may be transmitted over the telephone network if
   a. Their speed is low enough
   b. They are converted to analog first
   c. They are ac instead of dc
   d. They are digital only

11. Start and stop bits, respectively, are
   a. Mark, space
   b. Space, mark
   c. Space, space
   d. Mark, mark

12. Which of the following is correct?
   a. The bit rate may be greater than the baud rate
   b. The baud rate may be greater than the bit rate
   c. The bit and baud rates are always the same
   d. The bit and baud rates are not related

13. A modem converts
   a. Analog signals to digital
   b. Digital signals to analog
   c. Both a and b
   d. None of the above

14. Slow-speed modems use
   a. FSK
   b. BPSK
   c. QPSK
   d. QAM

15. A carrier recovery circuit is not needed with
    a. BPSK
    b. QPSK
    c. DPSK
    d. QAM

16. The basic modulator and demodulator circuits in PSK are
    a. PLLs
    b. Balanced modulators
    c. Shift registers
    d. Linear summers

17. The carrier used with a BPSK demodulator is
    a. Generated by oscillator
    b. The BPSK signal itself
    c. Twice the frequency of the transmitted carrier
    d. Recovered from the BPSK signal

18. A 9600 baud rate signal can pass over the voice-grade telephone line if which kind of modulation is used?
    a. BPSK
    b. QPSK
    c. DPSK
    d. QAM

19. Quadrature amplitude modulation is
    a. AM only
    b. QPSK only
    c. AM plus QPSK
    d. AM plus FSK

20. A QAM modulator does not use a(n)
    a. XNOR
    b. Bit splitter
    c. Balanced modulator
    d. 2-to-4 level converter

21. A rule or procedure that defines how data is to be transmitted is called a(n)
    a. Handshake
    b. Error-detection scheme
    c. Data specification
    d. Protocol

22. A popular PC protocol is
    a. Parity
    b. X modem
    c. CRC
    d. LRC

23. A synchronous transmission usually begins with which character?
    a. SYN
    b. STX
    c. SOH
    d. ETB

24. The characters making up the message in a synchronous transmission are collectively referred to as a data
    a. Set
    b. Sequence
    c. Block
    d. Collection

25. Bit errors in data transmission are usually cause by
    a. Equipment failures
    b. Typing mistake
    c. Noise
    d. Poor S/N ratio at the receiver

26. Which of the following is not a commonly used method of error detection?
    a. Parity
    b. BCC
    c. CRC
    d. Redundancy

27. Which of the following words has the correct parity bit? Assume odd parity. The last bit is the parity bit
    a. 11111111
    b. 11001101
    c. 00110101
    d. 00000000

28. Another name for parity is
    a. Vertical redundancy check
    b. Block check character
    c. Longitudinal redundancy check
    d. Cyclical redundancy check

29. Ten bit errors occur in two million transmitted. The bit error rate is
    a. 2×10^{-5}
    b. 5×10^{-5}
    c. 5×10^{-6}
    d. 2×10^{-6}

30. The building block of a parity or BCC generator is a(n)
    a. Shift register
    b. XOR
    c. 2 to 4 level converter
    d. UART

31. A longitudinal redundancy check produces a(n)
    a. Block check character
    b. Parity bit
    c. CRC
    d. Error correction

32. Dividing the data block by a constant produces a remainder that is used for error detection. It is called the
    a. Vertical redundancy check
    b. Horizontal redundancy check
    c. Block check character
    d. Cyclical redundancy check

33. A CRC generator uses which components?
    a. Balanced modulator
    b. Shift register
    c. Binary adder
    d. Multiplexer

34. Which of the following is not a LAN?
    a. PBX system
    b. Hospital system
    c. Office building system
    d. Cable TV system

35. The fastest LAN topology is the
    a. Ring
    b. Bus
    c. Star
    d. Square

36. Which is not a common LAN medium?
    a. Twin lead
    b. Twisted pair
    c. Fiber-optic cable
    d. Coax

37. A mainframe computer connected to multiple terminals and PCs usually uses which configuration?
    a. Bus
    b. Ring
    c. Star
    d. Tree

38. A small telephone switching system that can be used as a LAN is called a
    a. Ring
    b. WAN
    c. UART
    d. PBX

39. Which medium is the least susceptible to noise?
    a. Twin Lead
    b. Twisted pair
    c. Fiber optic cable
    d. Coax

40. Which medium is the most widely used in LANs?
    a. Twin Lead
    b. Twisted pair
    c. Fiber optic cable
    d. Coax

41. Transmitting the data signal directly over the medium is referred to as
    a. Baseband
    b. Broadband
    c. Ring
    d. Bus

42. The techniques of using modulation and FDM to transmit multiple data channels of a common medium is known as
    a. Baseband
    b. Broadband
    c. Ring
    d. Bus

43. What is the minimum bandwidth required to transmit a 56 kbits/s binary signal with no noise?
    a. 14 kHz
    b. 28 kHz
    c. 56 kHz
    d. 112 kHz

44. Sixteen different levels (symbols) are used to encode binary data. The channel bandwidth is 36 MHz. The maximum channel capacity is
    a. 18 Mbits/s
    b. 72 Mbits/s
    c. 288 Mbits/s
    d. 2.176 Gbits/s

45. What is the bandwidth required to transmit at a rate of 10Mbits/s in the presence of a 28-db S/N ratio?
    a. 1.075 MHz
    b. 5 MHz
    c. 10 MHz
    d. 10.75 MHz

46. Which circuit is common to bot frequency-hopping and direct-sequence SS transmitters?
    a. Correlator
    b. PSN code generator
    c. Frequency synthesizer
    d. Sweep generator

47. Spread spectrum stations sharing a band are identified by and distinguished from one another by
    a. PSN code
    b. Frequency of operation
    c. Clock rate
    d. Modulation type

48. The type of modulation most often used with direct-sequence SS is
    a. QAM
    b. SSB
    c. FSK
    d. PSK

49. The main circuit in a PSN generator is a(n)
    a. XOR
    b. Multiplexer
    c. Shift register
    d. Mixer

50. To a conventional narrowband receiver, an SS signal appears to be like
    a. Noise
    b. Fading
    c. A jamming signal
    d. An intermittent connection

51. Which of the following is not a benefit of SS?
    a. Jam-proof
    b. Security
    c. Immunity to fading
    d. Noise proof

52. Spread spectrum is a form of multiplexing
    a. True
    b. False

53. The most critical and difficult part of receiving a direct-sequence SS signal is
    a. Frequency synthesis
    b. Synchronism
    c. PSN code generation
    d. Carrier recovery

COMMUNICATION ELECTRONICS
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Chapter 13: Fiber Optic Communications

1. Which of the following is not a common application of fiber-optic cable?
    a. Closed circuit TV
    b. Consumer network
    c. Long - distance telephone systems
    d. Consumer TV

2. Total internal reflection takes place if the light ray strikes the interface at an angle with what relationship to the critical angle?
    a. Less than
    b. Greater than
    c. Equal to
    d. Zero

3. The operation of a fiber-optic cable is based on the principle of
    a. Refraction
    b. Reflection
    c. Dispersion
    d. Absorption

4. Which of the following is not a common type of fiber-optic cable?
    a. Single - mode step - index
    b. Multimode graded - index
    c. Single - mode grade - index
    d. Multimode step - index

5. Cable attenuation is usually expressed in terms of
    a. Loss per foot
    b. **dB/km**
    c. Intensity per mile
    d. Voltage drop per inch

6. Which cable length has the highest attenuation?
    a. 1 km
    b. **2 km**
    c. 95 ft
    d. 500 ft

7. The upper pulse rate and information-carrying capacity of a cable is limited by
    a. Pulse shortening
    b. **Attenuation**
    c. Light leakage
    d. Modal dispersion

8. The core of a fiber-optic cable is made of
    a. Air
    b. Glass
    c. Diamond
    d. Quartz

9. The core of a fiber-optic cable is surrounded by
    a. Wire braid shield
    b. Kevlar
    c. Cladding
    d. Plastic insulation

10. The speed of light in plastic compared to the speed of light in air is
    a. Less
    b. More
    c. The same
    d. Zero

11. Which of the following is not a major benefit of fiber-optic cable?
    a. Immunity from interference
    b. Excellent data security
    c. No electrical safety problems
    d. Lower cost

12. The main benefit of light-wave communications over microwaves or any other communications media are
    a. Lower cost
    b. Better security
    c. Wider bandwidth
    d. Freedom from interference

13. Which of the following is not part of the optical spectrum
    a. Infrared
    b. Ultraviolet
    c. Visible color
    d. X-rays

14. The wavelength of visible light extends from
    a. 0.8 to 1.6 μm
    b. 400 to 750 nm
    c. 200 to 600 nm
    d. 700 to 1200 nm

15. The speed of light is
    a. 180,000 mi/h
    b. 300,000 mi/h
    c. 300,000 m/s
    d. 300,000,000 m/s

16. Refraction is the
    a. Bending of light waves
    b. Reflection of light waves
    c. Distortion of light waves
    d. Diffusion of light waves

17. The ratio of the speed of light in air to the speed of light in another substance is called the
    a. Speed factor
    b. Index of reflection
    c. Index of refraction
    d. Speed gain

18. A popular light wavelength in fiber-optic cable is
    a. 0.7 μm
    b. 1.3 μm
    c. 1.5 μm
    d. 1.8 μm

19. Which type of fiber-optic cable is the most widely used?
    a. Single - mode step - index
    b. Multimode step-index
    c. Single - mode graded - index
    d. Multimode graded - index

20. Which type of fiber-optic cable is best for very high speed data?
    a. Single - mode step - index
    b. Multimode step-index
    c. Single - mode graded - index
    d. Multimode graded - index

21. Which type of fiber-optic cable has the least modal dispersion?
    a. Single - mode step - index
    b. Multimode step-index
    c. Single - mode graded - index
    d. Multimode graded - index

22. Which of the following is not a factor in cable light loss?
    a. Reflection
    b. Absorption
    c. Scattering
    d. Dispersion

23. A distance of 8 km is the same as
    a. 2.5 mi
    b. 5 mi
    c. 8 mi
    d. 12.9 mi

24. A fiber-optic cable has a loss of 15 db/km. The attenuation in a cable 1000ft long is
    a. 4.57 dB
    b. 9.3 dB
    c. 24 dB
    d. 49.2 dB

25. Fiber-optic with attenuations of 1.8, 3.4, 5.9 and 18 dB are linked together. The total loss is
    a. 7.5 dB
    b. 19.8 dB
    c. 29.1 dB
    d. 650 dB

26. Which light emitter is preferred for high-speed data in a fiber-optic system?
    a. Incandescent
    b. LED
    c. Neon
    d. Laser

27. Most fiber-optic light sources emit light in which spectrum?
    a. Visible
    b. Infrared
    c. Ultraviolet
    d. X- ray

28. Both LEDs and ILDs operate correctly with
    a. Forward bias
    b. Reverse bias
    c. Neither a or b
    d. Either a or b

29. Single-frequency light is called
    a. Pure
    b. Intense
    c. Coherent
    d. Monochromatic

30. Laser light is very bright because it is
    a. Pure
    b. White
    c. Coherent
    d. Monochromatic

31. Which of the following is not a common detector?
    a. PIN diode
    b. Photovoltaic diode
    c. Photodiode
    d. Avalanche photodiode

32. Which of the following is the fastest light sensor?
    a. PIN diode
    b. Photovoltaic diode
    c. Phototransistor
    d. Avalanche photodiode

33. Photodiodes operate properly with
    a. Forward bias
    b. Reverse bias
    c. Neither a or b
    d. Either a or b

34. The product of the bit rate and distance of a fiber-optic system is 2Gbits-km/s. What is the maximum rate at 5 km?
    a. 100 Mbits/s
    b. 200 Mbits/s
    c. 400 Mbits/s
    d. 1000 Mbits/s

35. Which fiber-optic system is better?
    a. 3 repeaters
    b. 8 repeaters
    c. 11 repeaters
    d. 20 repeaters

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Chapter 14: Modern Communication Applications

1. Printed documents to be transmitted by fax are converted into a baseband electrical signal by the process of
    a. Relfelction
    b. Scanning
    c. Modulation
    d. Light Variation

2. most commonly used light sensor in a modern fax machine is a
    a. Phototube
    b. Phototransistor
    c. Liquid - crystal display
    d. Charge coupled device

3. In FM fax, the frequencies for black and white are
    a. 1500 and 2300 Hz
    b. 2300 and 1500 Hz
    c. 1300 and 2400 Hz
    d. 1070 and 1270 Hz

4. Which resolution produces the best quality fax?
    a. 96 lines per inch
    b. 150 lines per inch
    c. 200 lines per inch
    d. 400 lines per inch

5. Group 2 fax uses which modulation?
    a. SSB
    b. FSK
    c. Vestigial sideband AM
    d. PSK

6. The most widely used fax standard is
    a. Group 1
    b. Group 2
    c. Group 3
    d. Group 4

7. Group 3 fax uses which modulation?
    a. QAM
    b. FSK
    c. Vestigial sideband AM
    d. FM

8. Most fax printers are of which type?
    a. Impact
    b. Thermal
    c. Electrosensitive
    d. Laser xerographic

9. Facsimile standards are set by the
    a. FCC
    b. DOD
    c. CCITT
    d. IEEE

10. What type of graphics is commonly transmitted by radio fax?
    a. Newspaper text
    b. Architectural drawings
    c. Cable movies
    d. Satellite weather photos

11. The transmission speed of group 4 fax is
    a. 4800 baud
    b. 9600 baud
    c. 56 kbits/s
    d. 192 kbits/s

12. The master control center for a cellular telephone system is the
    a. Cell site
    b. Mobile telephone switching office
    c. Center office
    d. Branch office

13. Each cell site contains a
    a. Repeater
    b. Control computer
    c. Direct link to a branch exchange
    d. Touch - tone processor

14. Multiple cells within an area may use the same channel frequencies
    a. True
    b. False

15. Cellular telephones use which type of operation?
    a. Simplex
    b. Half - duplex
    c. Full-duplex
    d. Triplex

16. The maximum frequency deviation of an FM cellular transmitter is
    a. 6 kHz
    b. 12 kHz
    c. 30 kHz
    d. 45 kHz

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17. The maximum output power of a cellular transmitter is
    a. 4.75 W
    b. 1.5 W
    c. 3 W
    d. 5 W

18. Receive channel 22 is 870.66 MHz. Receive channel 23 is
    a. 870.36 MHz
    b. 870.63 MHz
    c. 870.96 MHz
    d. 870.69 MHz

19. A transmit channel has a frequency of 837.6 MHz. The receive channel frequency is
    a. 729.6 MHz
    b. 837.6 MHz
    c. 867.6 MHz
    d. 882.6 MHz

20. A receive channel frequency is 872.4 MHz. To develop an 82.2 MHz IF, the frequency synthesizer must supply an LO signal of
    a. 790.2 MHz
    b. 827.4 MHz
    c. 954.6 MHz
    d. 967.4 MHz

21. The output power of a cellular radio is controlled by the
    a. User or Caller
    b. Cell site
    c. Caller party
    d. MTSO

22. When the signal from a mobile cellular unit drops below a certain level, what action occurs?
    a. The unit is "handed off" to a closer cell.
    b. The call is terminated.
    c. The MTSO increases power level.
    d. The cell site switches antennas.

23. In a cellular radio, the duplexer is a
    a. Ferrite isolator
    b. Wavegiude assembly
    c. Pair of TR/ ATR tubes
    d. Pair of sharp bandpass filter

24. The time from the transmission of a radar pulse to its reception is 0.12 ms. The distance to the target is how many nautical miles?
    a. 4.85 nmi
    b. 9.7 nmi
    c. 11.2 nmi
    d. 18.4 nmi

25. The ability of a radar to determine the bearing to a target depends upon the
    a. Antenna directivity
    b. Speed of light
    c. Speed of the target
    d. Frequency of the signal

26. The pulse duration of a radar signal is 600 ns. The PRF is 185 pulses per second. The duty cycle is
    a. 1.1 %
    b. 5.5 %
    c. 31 %
    d. 47 %

27. The Doppler effect is used to produce modulation of which type of radar signal?
    a. Pulse
    b. CW

28. The Doppler Effect allows which characteristics of a target to be measured?
    a. Distance
    b. Azimuth
    c. Altitude
    d. Speed

29. The Doppler Effect is a change in what signal characteristic produced by relative motion between the radar set and a target?
    a. Amplitude
    b. Phase
    c. Frequency
    d. Duty cycle

30. The most widely used radar transmitter component is a
    a. Klystron
    b. Magnetron
    c. TWT
    d. Power transistor

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31. Low-power radar transmitters and receiver LOs use which component?
   a. GaAs FET
   b. Magnetron
   c. Gunn diode
   d. Klystron

32. What component in a duplexer protects the receiver from the high-power transmitter output?
   a. Waivegiude
   b. Bandpass filter
   c. Notch filter
   d. Spark gap

33. Most radar antennas usa a
   a. Dipole
   b. Broadside array
   c. Horn and parabolic reflector
   d. Collinear array

34. The most common radar display is the
   a. A scan.
   b. Color CRT
   c. Liquid - crystal display
   d. Plan position indicator

35. A radar antenna using multiple dipoles or slot antennas in a matrix with variable phase shifters is called a(n)
   a. A scan
   b. Phased array
   c. Broadside
   d. Circular polarized array

36. Police radars use which technique?
   a. Pulse
   b. CW

37. Which of the following is a typical radar operating frequency?
   a. 60 MHz
   b. 450 MHz
   c. 900 MHz
   d. 10 GHz

38. The TV signal uses which types of modulation for picture and sound respectively?
   a. AM, FM
   b. DSB , FM
   c. FM, AM
   d. AM, DSB

39. If a TV sound transmitter has a carrier frequency of 197.75 MHz, the picture carrier is
   a. 191.75 MHz
   b. 193.25 MHz
   c. 202.25 MHz
   d. 203.75 MHz

40. The total bandwidth of an NTSC TV signal is
   a. 3.58 MHz
   b. 4.5 MHz
   c. 6 MHz
   d. 10.7 MHz

41. What is the total number of interlaced scan lines in one complete frame of a NTSC U.S. TV signal?
   a. 262 ½
   b. 525
   c. 480
   d. 625

42. What keeps the scanning process at the receiver in step with the scanning in the picture tube at receiver?
   a. Nothing
   b. Color burst
   c. Sync pulses
   d. Deflection oscillators

43. What is the black-and-white or monochrome brightness signal in TV called
   a. RGB
   b. Color subcarrier
   c. Q and I
   d. Luminance Y

44. What is the name of the solid-state imaging device used int TV cameras that converts the light in a scene into an electrical signal?
   a. CCD
   b. Photodiode matrix
   c. Vidicon
   d. MOSFET array

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45. The I and Q composite color signals are multiplexed onto the picture carrier by modulating a 3.58 MHz subcarrier using
    a. FM
    b. PM
    c. DSB AM
    d. Vestigial sideband AM

46. The assembly around the neck of a picture tube that produces the magnetic fields that deflect and scan the electron beams is called the
    a. Shadow mask
    b. Phosphor
    c. Electron gun
    d. Yoke

47. The picture and sound carrier frequencies in a TV receiver IF are respectively
    a. 41.25 and 45.75 MHz
    b. 45.75 and 41.25 MHz
    c. 41.75 and 45.25 MHz
    d. 45.25 and 41.75 MHz

48. The sound IF in a TV receiver is
    a. 4.5 MHz
    b. 10.7 MHz
    c. 41.25 MHz
    d. 45.75 MHz

49. What type of circuit is used to modulate and demodulate the color signals?
    a. Phased - locked loop
    b. Differential peak detector
    c. Quadrature detector
    d. Balanced demodulator

50. What circuit in the TV receiver is used to develop the high voltage needed to operate the picture tube?
    a. Low - voltage power supply
    b. Horizontal output
    c. Vertical sweep
    d. Sync separator

51. What ensures proper color synchronization at the receiver?
    a. Sync pulses
    b. Quadrature modulation
    c. 4.5 MHz carrier spacing
    d. 3.58 MHz color burst

52. Which of the following is not a benefit of cable TV?
    a. Lower-cost reception
    b. Greater reliability
    c. Less noise, stronger signals
    d. Premium cable channels

53. What technique is used to permit hundreds of TV signals to share a common cable?
    a. Frequency modulation
    b. Mixing
    c. Frequency division multiplexing
    d. Time division multiplexing